Network Lab - Tianyu Chen

Introduction

This project presents the design and implementation of a multi-site enterprise network connecting two separate offices. The overall network topology is shown in **Figure 1**.

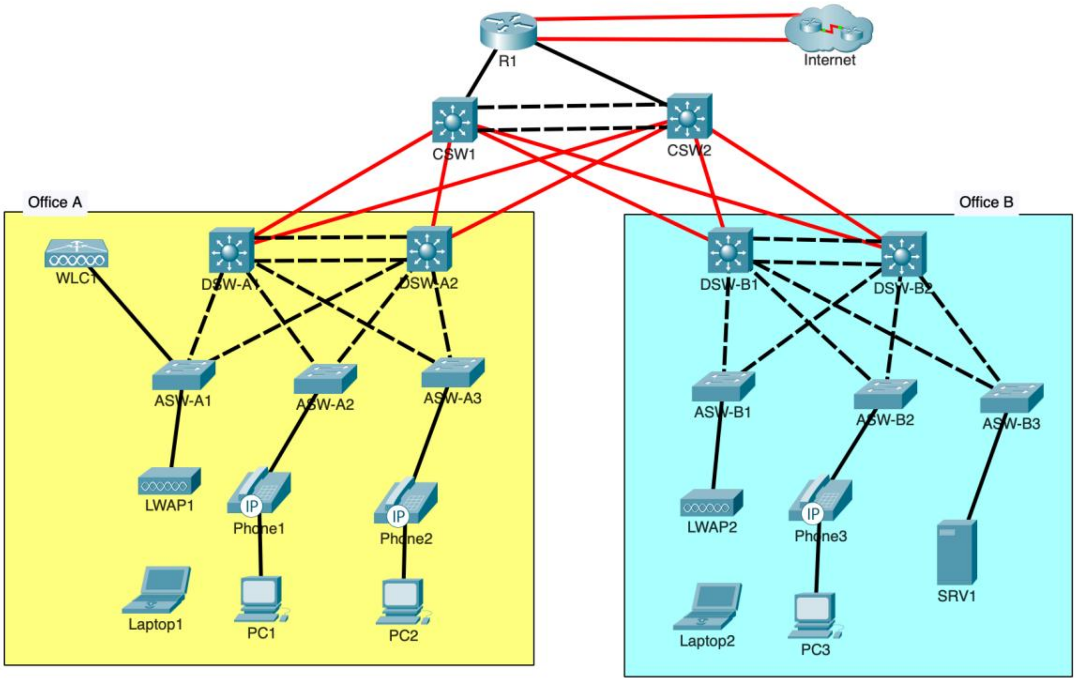

Figure 1. Network Topology

The project covers a range of network technologies and configurations, including **VLANs**, **Layer 2/3 EtherChannel**, **RSTP**, **HSRP**, **IP addressing**, and **static and dynamic routing**. These technologies are used to provide network segmentation, redundancy, high availability, and reliable connectivity between different network segments.

The lab also implements essential network services, including **DHCP**, **DNS**, **NTP**, **SNMP**, **Syslog**, **FTP**, **SSH**, **and NAT**. Network security is addressed through **Access Control Lists (ACLs)** and various **Layer 2 security mechanisms**. In addition, the project includes **IPv6** and **wireless network configuration** to support different network environments and communication requirements.

The following sections present the key configuration commands and relevant verification results for each stage of the implementation. Rather than providing complete configuration files, this document focuses on the **network design**, **implementation process**, and **key technologies and configurations** used to achieve the required functionality.

Part 1 - Initial setup

1.Configure the appropriate hostname on each router/switch.
2.Configure the enable secret **cisco** on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
3.Configure the user account **cisco** with secret **cisco** on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
4.Configure the console line to require login with a local user account. Set a 30-minute inactivity timeout. Enable synchronous logging.
Switch#conf t
Switch(config)#hostname CSW1
CSW1(config)#enable algorithm-type scrypt secret cisco
CSW1(config)#username cisco algorithm-type scrypt secret cisco
CSW1(config)#line console 0
CSW1(config-line)#login local
CSW1(config-line)#exec-timeout 30
CSW1(config-line)#logging synchronous
CSW1(config-line)#end
CSW1#write memory

Part 2 – VLANs, Layer-2 EtherChannel

1. In Office A, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-A1 and DSW-A2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel.
DSW-A2#conf t
DSW-A2(config)#interface range g1/0/4-5
DSW-A2(config-if-range)#channel-group 1 mode desirable
DSW-A2(config-if-range)#end

2. In Office B, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-B1 and DSW-B2 using an open standard protocol. Both switches should actively try to form an EtherChannel.
DSW-B2#conf t
DSW-B2(config)#interface range g1/0/4-5
DSW-B2(config-if-range)#channel-group 1 mode active
DSW-B2(config-if-range)#end

3.Configure all links between Access and Distribution switches, including the EtherChannels, as trunk links.

- **a.** Explicitly disable DTP on all ports.
- **b.** Set each trunk's native VLAN to VLAN 1000 (unused).
- **c.** In Office A, allow VLANs 10, 20, 40, and 99 on all trunks.
- **d.** In Office B, allow VLANs 10, 20, 30, and 99 on all trunks.

DSW-A2(config)#interface range g1/0/1-3
DSW-A2(config-if-range)#switchport mode trunk
DSW-A2(config-if-range)#switchport nonegotiate
DSW-A2(config-if-range)#switchport trunk native vlan 1000
DSW-A2(config-if-range)#switchport trunk allowed vlan 10,20,40,99
DSW-A2(config-if-range)#exit

DSW-A2(config)#interface port-channel 1
DSW-A2(config-if)#switchport mode trunk
DSW-A2(config-if)#switchport nonegotiate
DSW-A2(config-if)#switchport trunk native vlan 1000
DSW-A2(config-if)#switchport trunk allowed vlan 10,20,40,99
DSW-A2(config-if)#end

4.Configure one of each office's Distribution switches as a VTPv2 server. Use domain name **Cisco**.

a. Verify that other switches join the domain.
b. Configure all Access switches as VTP clients.

DSW-A1(config)#vtp version 2
DSW-A1(config)#vtp domain Cisco
DSW-A1(config)#vtp mode server

5.In Office A, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.

a. VLAN 10: PCs
b. VLAN 20: Phones
c. VLAN 40: Wi-Fi
d. VLAN 99: Management

DSW-A1(config)#vlan 10
DSW-A1(config-vlan)#name PCs
DSW-A1(config-vlan)#vlan 20
DSW-A1(config-vlan)#name Phones
DSW-A1(config-vlan)#vlan 40
DSW-A1(config-vlan)#name Wi-Fi
DSW-A1(config-vlan)#vlan 99
DSW-A1(config-vlan)#name Management

6. In Office B, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.

a. VLAN 10: PCs
b. VLAN 20: Phones
c. VLAN 30: Servers
d. VLAN 99: Management

DSW-B1(config)#vlan 10
DSW-B1(config-vlan)#name PCs
DSW-B1(config-vlan)#vlan 20
DSW-B1(config-vlan)#name Phones
DSW-B1(config-vlan)#vlan 30
DSW-B1(config-vlan)#name Servers
DSW-B1(config-vlan)#vlan 99
DSW-B1(config-vlan)#name Management

7.Configure each Access switch's access port.

a. LWAPs will not use FlexConnect
b. PCs in VLAN 10, Phones in VLAN 20
c. SRV1 in VLAN 30
d. Manually configure access mode and explicitly disable DTP

ASW-A2(config)#interface f0/1

ASW-A2(config-if)#switchport mode access
ASW-A2(config-if)#switchport nonegotiate
ASW-A2(config-if)#switchport access vlan 10
ASW-A2(config-if)#switchport voice vlan 20
ASW-A2(config-if)#end

8.Configure ASW-A1's connection to WLC1:

a. It must support the Wi-Fi and Management VLANs.
b. The Management VLAN should be untagged.
c. Disable DTP.

ASW-A1(config)#interface range f0/2
ASW-A1(config-if-range)#switchport mode trunk
ASW-A1(config-if-range)#switchport trunk allowed vlan 40,99
ASW-A1(config-if-range)#switchport trunk native vlan 99
ASW-A1(config-if-range)#switchport nonegotiate
ASW-A1(config-if-range)#exit

9. Administratively disable all unused ports on Access and Distribution switches.
ASW-A1#conf t
ASW-A1(config)#interface range f0/3-24
ASW-A1(config-if-range)#shutdown
ASW-A1(config-if-range)#end

Part 3 – IP Addresses, Layer-3 EtherChannel, HSRP

1.Configure the following IP addresses on R1's interfaces and enable them:

a. G0/0/0: DHCP client
b. G0/1/0: DHCP client
c. G0/0: 10.0.0.33/30
d. G0/1: 10.0.0.37/30
e. Loopback0: 10.0.0.76/32

R1(config)#interface g0/0/0
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown
R1(config-if)#interface g0/1/0
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface g0/0
R1(config-if)#ip address 10.0.0.33 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#interface g0/1
R1(config-if)#ip address 10.0.0.37 255.255.255.252
R1(config-if)#no shutdown
R1(config)#interface loopback 0
R1(config-if)#ip address 10.0.0.76 255.255.255.255
R1(config-if)#exit

2.Enable IPv4 routing on all Core and Distribution switches.
DSW-B2#conf t

DSW-B2(config)#ip routing
DSW-B2(config)#end

3.Create a Layer-3 EtherChannel between CSW1 and CSW2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel. Configure the following IP addresses:

a. CSW1 PortChannel1: 10.0.0.41/30
b. CSW2 PortChannel1: 10.0.0.42/30

CSW2(config)#interface range g1/0/2-3
CSW2(config-if-range)#channel-group 1 mode desirable
CSW2(config-if-range)#exit
CSW2(config)#interface port-channel 1
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.42 255.255.255.252
CSW2(config-if)#exit

4. Configure the following IP addresses on CSW1. Disable all unused interfaces.

a. G1/0/1: 10.0.0.34/30
b. G1/1/1: 10.0.0.45/30
c. G1/1/2: 10.0.0.49/30
d. G1/1/3: 10.0.0.53/30
e. G1/1/4: 10.0.0.57/30
f. Loopback0: 10.0.0.77/32

CSW1(config)#interface g1/0/1
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.34 255.255.255.252
CSW1(config-if)#exit
CSW1(config)#interface g1/1/1
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.45 255.255.255.252
CSW1(config-if)#exit
CSW1(config)#interface g1/1/2
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.49 255.255.255.252
CSW1(config-if)#exit
CSW1(config)#interface g1/1/3
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.53 255.255.255.252
CSW1(config-if)#exit
CSW1(config)#interface g1/1/4
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.57 255.255.255.252
CSW1(config-if)#exit
CSW1(config)#interface loopback 0
CSW1(config-if)#ip address 10.0.0.77 255.255.255.255
CSW1(config-if)#exit
CSW1(config)#interface range g1/0/4-24
CSW1(config-if-range)#shutdown
CSW1(config-if-range)#exit

5.Configure the following IP addresses on CSW2. Disable all unused interfaces.

a. G1/0/1: 10.0.0.38/30
b. G1/1/1: 10.0.0.61/30
c. G1/1/2: 10.0.0.65/30
d. G1/1/3: 10.0.0.69/30
e. G1/1/4: 10.0.0.73/30
f. Loopback0: 10.0.0.78/32

CSW2(config)#interface g1/0/1
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.38 255.255.255.252
CSW2(config-if)#exit
CSW2(config)#interface g1/1/1
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.61 255.255.255.252
CSW2(config-if)#exit
CSW2(config)#interface g1/1/2
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.65 255.255.255.252
CSW2(config-if)#exit
CSW2(config)#interface g1/1/3
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.69 255.255.255.252
CSW2(config-if)#exit
CSW2(config)#interface g1/1/4
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.73 255.255.255.252
CSW2(config-if)#exit
CSW2(config)#interface loopback 0
CSW2(config-if)#ip address 10.0.0.78 255.255.255.255
CSW2(config-if)#end
CSW2(config)#interface range g1/0/4-24
CSW2(config-if-range)#shutdown
CSW2(config-if-range)#end

6.Configure the following IP addresses on DSW-A1:

a. G1/1/1: 10.0.0.46/30
b. G1/1/2: 10.0.0.62/30
c. Loopback0: 10.0.0.79/32

DSW-A1(config)#interface g1/1/1
DSW-A1(config-if)#no switchport
DSW-A1(config-if)#ip address 10.0.0.46 255.255.255.252
DSW-A1(config-if)#exit
DSW-A1(config)#interface g1/1/2
DSW-A1(config-if)#no switchport
DSW-A1(config-if)#ip address 10.0.0.62 255.255.255.252
DSW-A1(config-if)#exit
DSW-A1(config)#interface loopback 0
DSW-A1(config-if)#ip address 10.0.0.79 255.255.255.255
DSW-A1(config-if)#exit

7.Configure the following IP addresses on DSW-A2:

a. G1/1/1: 10.0.0.50/30

b. G1/1/2: 10.0.0.66/30

c. Loopback0: 10.0.0.80/32

DSW-A2(config)#interface g1/1/1
DSW-A2(config-if)#no switchport
DSW-A2(config-if)#ip address 10.0.0.50 255.255.255.252
DSW-A2(config-if)#exit
DSW-A2(config)#interface g1/1/2
DSW-A2(config-if)#no switchport
DSW-A2(config-if)#ip address 10.0.0.66 255.255.255.252
DSW-A2(config-if)#exit
DSW-A2(config)#interface loopback 0
DSW-A2(config-if)#ip address 10.0.0.8 255.255.255.255
DSW-A2(config-if)#exi

8.Configure the following IP addresses on DSW-B1:

a. G1/1/1: 10.0.0.54/30

b. G1/1/2: 10.0.0.70/30

c. Loopback0: 10.0.0.81/32

DSW-B1(config)#interface g1/1/1
DSW-B1(config-if)#no switchport
DSW-B1(config-if)#ip address 10.0.0.54 255.255.255.252
DSW-B1(config-if)#exit
DSW-B1(config)#interface g1/1/2
DSW-B1(config-if)#no switchport
DSW-B1(config-if)#ip address 10.0.0.70 255.255.255.252
DSW-B1(config-if)#exit
DSW-B1(config)#interface loopback 0
DSW-B1(config-if)#ip address 10.0.0.81 255.255.255.255
DSW-B1(config-if)#end

9.Configure the following IP addresses on DSW-B2:

a. G1/1/1: 10.0.0.58/30

b. G1/1/2: 10.0.0.74/30

c. Loopback0: 10.0.0.82/32

DSW-B2#conf t
DSW-B2(config)#interface g1/1/1
DSW-B2(config-if)#no switchport
DSW-B2(config-if)#ip address 10.0.0.58 255.255.255.252
DSW-B2(config-if)#exit
DSW-B2(config)#interface g1/1/2
DSW-B2(config-if)#no switchport
DSW-B2(config-if)#ip address 10.0.0.74 255.255.255.252
DSW-B2(config-if)#exit
DSW-B2(config)#interface loopback 0
DSW-B2(config-if)#ip address 10.0.0.82 255.255.255.255
DSW-B2(config-if)#exit

10. Manually configure SRV1's IP settings:

a. Default Gateway: 10.5.0.1
b. IPv4 Address: 10.5.0.4
c. Subnet Mask: 255.255.255.0

11. Configure the following management IP addresses on the Access switches (interface VLAN 99), and configure the appropriate subnet's first usable address as the default gateway.

a. ASW-A1: 10.0.0.4/28
b. ASW-A2: 10.0.0.5/28
c. ASW-A3: 10.0.0.6/28
d. ASW-B1: 10.0.0.20/28
e. ASW-B2: 10.0.0.21/28
f. ASW-B3: 10.0.0.22/28

ASW-A1(config)#interface vlan 99
ASW-A1(config-if)#ip address 10.0.0.4 255.255.255.240
ASW-A1(config-if)#exit
ASW-A1(config)#ip default-gateway 10.0.0.1
ASW-A1(config)#end

ASW-B1(config)#interface vlan 99
ASW-B1(config-if)#ip address 10.0.0.20 255.255.255.240
ASW-B1(config-if)#exit
ASW-B1(config)#ip default-gateway 10.0.0.17
ASW-B1(config)#end

12. Configure HSRPv2 group 1 for Office A's Management subnet (VLAN 99). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

a. Subnet: 10.0.0.0/28
b. VIP: 10.0.0.1
c. DSW-A1: 10.0.0.2
d. DSW-A2: 10.0.0.3

DSW-A1(config)#interface vlan 99
DSW-A1(config-if)#ip address 10.0.0.2 255.255.255.240
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 1 ip 10.0.0.1
DSW-A1(config-if)#standby 1 priority 105
DSW-A1(config-if)#standby 1 preempt
DSW-A1(config-if)#end

DSW-A2(config)#interface vlan 99
DSW-A2(config-if)#ip address 10.0.0.3 255.255.255.240
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 1 ip 10.0.0.1
DSW-A2(config-if)#end

13. Configure HSRPv2 group 2 for Office A's PCs subnet (VLAN 10). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

a. Subnet: 10.1.0.0/24

b. VIP: 10.1.0.1
c. DSW-A1: 10.1.0.2
d. DSW-A2: 10.1.0.3

DSW-A1(config)#interface vlan 10
DSW-A1(config-if)#ip address 10.1.0.2 255.255.255.0
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 2 ip 10.1.0.1
DSW-A1(config-if)#standby 2 priority 105
DSW-A1(config-if)#standby 2 preempt
DSW-A1(config-if)#end

DSW-A2(config)#interface vlan 10
DSW-A2(config-if)#ip address 10.1.0.3 255.255.255.0
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 2 ip 10.1.0.1
DSW-A2(config-if)#end

14. Configure HSRPv2 group 3 for Office A's Phones subnet (VLAN 20). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

a. Subnet: 10.2.0.0/24
b. VIP: 10.2.0.1
c. DSW-A1: 10.2.0.2
d. DSW-A2: 10.2.0.3

DSW-A1(config)#interface vlan 20
DSW-A1(config-if)#ip address 10.2.0.2 255.255.255.0
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 3 ip 10.2.0.1
DSW-A1(config-if)#end

DSW-A2(config)#interface vlan 20
DSW-A2(config-if)#ip address 10.2.0.3 255.255.255.0
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 3 ip 10.2.0.1
DSW-A2(config-if)#standby 3 priority 105
DSW-A2(config-if)#standby 3 preempt
DSW-A2(config-if)#end

15. Configure HSRPv2 group 4 for Office A's Wi-Fi subnet (VLAN 40). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

a. Subnet: 10.6.0.0/24
b. VIP: 10.6.0.1
c. DSW-A1: 10.6.0.2
d. DSW-A2: 10.6.0.3

DSW-A1(config)#interface vlan 40
DSW-A1(config-if)#ip address 10.6.0.2 255.255.255.0
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 4 ip 10.6.0.1
DSW-A1(config-if)#end

DSW-A2(config)#interface vlan 40
DSW-A2(config-if)#ip address 10.6.0.3 255.255.255.0
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 4 ip 10.6.0.1
DSW-A2(config-if)#standby 4 priority 105
DSW-A2(config-if)#standby 4 preempt
DSW-A2(config-if)#end

16. Configure HSRPv2 group 1 for Office B's Management subnet (VLAN 99). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

   a. Subnet: 10.0.0.16/28
   b. VIP: 10.0.0.17
   c. DSW-B1: 10.0.0.18
   d. DSW-B2: 10.0.0.19

DSW-B1(config)#interface vlan 99
DSW-B1(config-if)#ip address 10.0.0.18 255.255.255.240
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 1 ip 10.0.0.17
DSW-B1(config-if)#standby 1 priority 105
DSW-B1(config-if)#standby 1 preempt
DSW-B1(config-if)#end

DSW-B2(config)#interface vlan 99
DSW-B2(config-if)#ip address 10.0.0.19 255.255.255.240
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 1 ip 10.0.0.17
DSW-B2(config-if)#end

17. Configure HSRPv2 group 2 for Office B's PCs subnet (VLAN 10). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

   a. Subnet: 10.3.0.0/24
   b. VIP: 10.3.0.1
   c. DSW-B1: 10.3.0.2
   d. DSW-B2: 10.3.0.3

DSW-B1(config)#interface vlan 10
DSW-B1(config-if)#ip address 10.3.0.2 255.255.255.0
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 2 ip 10.3.0.1
DSW-B1(config-if)#standby 2 priority 105
DSW-B1(config-if)#standby 2 preempt
DSW-B1(config-if)#end

DSW-B2(config)#interface vlan 10
DSW-B2(config-if)#ip address 10.3.0.3 255.255.255.0
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 2 ip 10.3.0.1
DSW-B2(config-if)#end

18. Configure HSRPv2 group 3 for Office B's Phones subnet (VLAN 20). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

  a. Subnet: 10.4.0.0/24
  b. VIP: 10.4.0.1
  c. DSW-B1: 10.4.0.2
  d. DSW-B2: 10.4.0.3

DSW-B1(config)#interface vlan 20
DSW-B1(config-if)#ip address 10.4.0.2 255.255.255.0
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 3 ip 10.4.0.1
DSW-B1(config-if)#end

DSW-B2(config)#interface vlan 20
DSW-B2(config-if)#ip address 10.4.0.3 255.255.255.0
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 3 ip 10.4.0.1
DSW-B2(config-if)#standby 3 priority 105
DSW-B2(config-if)#standby 3 preempt
DSW-B2(config-if)#end

19. Configure HSRPv2 group 4 for Office B's Servers subnet (VLAN 30). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

  a. Subnet: 10.5.0.0/24
  b. VIP: 10.5.0.1
  c. DSW-B1: 10.5.0.2
  d. DSW-B2: 10.5.0.3

DSW-B1(config)#interface vlan 30
DSW-B1(config-if)#ip address 10.5.0.2 255.255.255.0
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 4 ip 10.5.0.1
DSW-B1(config-if)#end

DSW-B2(config)#interface vlan 30
DSW-B2(config-if)#ip address 10.5.0.3 255.255.255.0
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 4 ip 10.5.0.1
DSW-B2(config-if)#standby 4 priority 105
DSW-B2(config-if)#standby 4 preempt
DSW-B2(config-if)#end

Part 4 – Rapid Spanning Tree Protocol

1.Configure Rapid PVST+ on all Access and Distribution switches.

  a. Ensure that the Root Bridge for each VLAN aligns with the HSRP Active router by configuring the lowest possible STP priority.
  b. Configure the HSRP Standby Router for each VLAN with an STP priority one increment above the lowest priority.

ASW-A1(config)#spanning-tree mode rapid-pvst
ASW-A1(config)#interface f0/2
ASW-A1(config-if)#spanning-tree portfast trunk
ASW-A1(config-if)#spanning-tree bpduguard enable
ASW-A1(config-if)#end

DSW-A2(config)#spanning-tree mode rapid-pvst
DSW-A2(config)#spanning-tree vlan 20,40 priority 0
DSW-A2(config)#spanning-tree vlan 10,99 priority 4096

2. Enable PortFast and BPDU Guard on all ports connected to end hosts (including WLC1). Perform the configurations in interface config mode.
ASW-A1(config)#interface f0/1
ASW-A1(config-if)#spanning-tree portfast
ASW-A1(config-if)#spanning-tree bpduguard enable

Part 5 – Static and Dynamic Routing

1.Configure OSPF on R1 (LAN-facing interfaces) and all Core and Distribution switches (all Layer-3 interfaces).

a. Use process ID 1 and Area 0.
b.Manually configure each device's RID to match the loopback interface IP.
c.On switches, use the network command to match the exact IP address of each interface.
d. On R1, enable OSPF in interface config mode.
e. Make sure OSPF is enabled on all loopback interfaces, too. Loopback interfaces should be passive.
f. Each Distribution switch's SVIs (except the Management VLAN SVI) should be passive, too.
g. Configure all physical connections between OSPF neighbors to use a network type that doesn't elect a DR/BDR. NOTE: This doesn't work on the Layer-3 PortChannel interfaces between CSW1/CSW2. Leave them as the default network type.

R1(config)#router ospf 1
R1(config-router)#router-id 10.0.0.76
R1(config-router)#passive-interface loopback 0
R1(config-router)#exit
R1(config)#interface loopback 0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#interface range g0/0-1
R1(config-if-range)#ip ospf 1 area 0
R1(config-if-range)#ip ospf network point-to-point
R1(config-if-range)#exit

CSW1(config)#router ospf 1
CSW1(config-router)#router-id 10.0.0.76
CSW1(config-router)#passive-interface loopback 0
CSW1(config-router)#passive-interface vlan 10
CSW1(config-router)#network 10.0.0.41 0.0.0.0 area 0

CSW1(config-router)#network 10.0.0.34 0.0.0.0 area 0
CSW1(config-router)#network 10.0.0.45 0.0.0.0 area 0
CSW1(config-router)#network 10.0.0.49 0.0.0.0 area 0
CSW1(config-router)#network 10.0.0.53 0.0.0.0 area 0
CSW1(config-router)#network 10.0.0.77 0.0.0.0 area 0
CSW1(config)#interface range g1/0/1, g1/1/1-4
CSW1(config-if-range)#ip ospf network point-to-point
CSW1(config-if-range)#exit

2.Configure one static default route for each of R1's Internet connections. They should be recursive routes.

a. Make the route via G0/1/0 a floating static route by configuring an AD value 1 greater than the default.

b. R1 should function as an OSPF ASBR, advertising its default route to other routers in the OSPF domain.

R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.1
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.5 2
R1(config)#router ospf 1
R1(config-router)#default-information originate
R1(config-router)#exit

Part 6 – Network Services: DHCP, DNS, NTP, SNMP, Syslog, FTP, SSH, NAT

1.Configure the following DHCP pools on R1 to make it serve as the DHCP server for hosts in Offices A and B. Exclude the first ten usable host addresses of each pool; they must not be leased to DHCP clients.

a. Pool: A-Mgmt
- **i.** Subnet: 10.0.0.0/28
- **ii.** Default gateway: 10.0.0.1
- **iii.** Domain name: cisco.com
- **iv.** DNS server: 10.5.0.4 (SRV1)
- **v.** WLC: 10.0.0.7

R1(config)#ip dhcp excluded-address 10.0.0.1 10.0.0.10
R1(config)#ip dhcp pool A-Mgmt
R1(dhcp-config)#network 10.0.0.0 255.255.255.240
R1(dhcp-config)#default-router 10.0.0.1
R1(dhcp-config)#domain-name cisco.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#option 43 ip 10.0.0.7
R1(dhcp-config)#end

b. Pool: A-PC
- **i.** Subnet: 10.1.0.0/24
- **ii.** Default gateway: 10.1.0.1
- **iii.** Domain name: cisco.com
- **iv.** DNS server: 10.5.0.4 (SRV1)

R1(config)#ip dhcp excluded-address 10.1.0.1 10.1.0.10
R1(config)#ip dhcp pool A-PC

R1(dhcp-config)#domain-name cisco.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#default-router 10.1.0.1
R1(dhcp-config)#end

c. Pool: A-Phone
- **i.** Subnet: 10.2.0.0/24
- **ii.** Default gateway: 10.2.0.1
- **iii.** Domain name: cisco.com
- **iv.** DNS server: 10.5.0.4 (SRV1)

R1(config)#ip dhcp excluded-address 10.2.0.1 10.2.0.10
R1(config)#ip dhcp pool A-Phone
R1(dhcp-config)#network 10.2.0.0 255.255.255.0
R1(dhcp-config)#domain-name cisco.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#default-router 10.2.0.1
R1(dhcp-config)#end

d. Pool: B-Mgmt
- **i.** Subnet: 10.0.0.16/28
- **ii.** Default gateway: 10.0.0.17
- **iii.** Domain name: cisco.com
- **iv.** DNS server: 10.5.0.4 (SRV1)
- **v.** WLC: 10.0.0.7

R1(config)#ip dhcp excluded-address 10.0.0.17 10.0.0.26
R1(config)#ip dhcp pool B-Mgmt
R1(dhcp-config)#network 10.0.0.16 255.255.255.240
R1(dhcp-config)#default-router 10.0.0.17
R1(dhcp-config)#domain-name cisco.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#option 43 ip 10.0.0.7
R1(dhcp-config)#end

e. Pool: B-PC
- **i.** Subnet: 10.3.0.0/24
- **ii.** Default gateway: 10.3.0.1
- **iii.** Domain name: cisco.com
- **iv.** DNS server: 10.5.0.4 (SRV1)

R1(config)#ip dhcp excluded-address 10.3.0.1 10.3.0.10
R1(config)#ip dhcp pool B-PC
R1(dhcp-config)#network 10.3.0.0 255.255.255.0
R1(dhcp-config)#domain-name cisco.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#default-router 10.3.0.1
R1(dhcp-config)#end

f. Pool: B-Phone
- **i.** Subnet: 10.4.0.0/24
- **ii.** Default gateway: 10.4.0.1
- **iii.** Domain name: cisco.com

iv. DNS server: 10.5.0.4 (SRV1)
R1(config)#ip dhcp excluded-address 10.4.0.1 10.4.0.10
R1(config)#ip dhcp pool B-Phone
R1(dhcp-config)#network 10.4.0.0 255.255.255.0
R1(dhcp-config)#domain-name cisco.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#default-router 10.4.0.1
R1(dhcp-config)#end

g. Pool: Wi-Fi
i. Subnet: 10.6.0.0/24
ii. Default gateway: 10.6.0.1
iii. Domain name: cisco.com
iv. DNS server: 10.5.0.4 (SRV1)
R1(config)#ip dhcp excluded-address 10.6.0.1 10.6.0.10
R1(config)#ip dhcp pool Wi-Fi
R1(dhcp-config)#network 10.6.0.0 255.255.255.0
R1(dhcp-config)#domain-name cisco.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#default-router 10.6.0.1
R1(dhcp-config)#end

2.Configure the Distribution switches to relay wired DHCP clients' broadcast messages to R1's Loopback0 IP address.
DSW-A1(config)#interface vlan 10
DSW-A1(config-if-range)#ip helper-address 10.0.0.76

3.Configure the following DNS entries on SRV1:
a. google.com = 172.253.62.100
b. youtube.com = 152.250.31.93
c. cisco.com = 66.235.200.145
d. www.cisco.com = cisco.com

4.Configure all routers and switches to use domain name **cisco.com** and use SRV1 as their DNS server.
CSW1(config)#ip domain-name cisco.com
CSW1(config)#ip name-server 10.5.0.1
CSW1(config)#exit

5. Configure NTP on R1:
a. Make R1 a stratum 5 NTP server.
b. R1 should learn the time from NTP server 216.239.35.0.
R1(config)#ntp master 5
R1(config)#ntp server 216.239.35.0
R1(config)#exit

6.All Core, Distribution, and Access switches should use R1's loopback interface as their NTP server.
a. Clients should authenticate R1 using key number **1** and the password **cisco**.

CSW1(config)#ntp authentication-key 1 md5 cisco
CSW1(config)#ntp trusted-key 1
CSW1(config)#ntp server 10.0.0.76 key 1
CSW1(config)#exit

7.Configure the SNMP community string **SNMPSTRING** on all routers and switches. The string should allow GET messages, but not SET messages.
R1(config)#snmp-server community SNMPSTRING ro
R1(config)#exit

8.Configure Syslog on all routers and switches:

> **a.** Send Syslog messages to SRV1. Messages of all severity levels should be logged.
> **b.** Enable logging to the buffer. Reserve 8192 bytes of memory for the buffer.

R1(config)#logging 10.5.0.4
R1(config)#logging trap debugging
R1(config)#logging buffered 8192
R1(config)#exit

9.Use FTP on R1 to download a new IOS version from SRV1:

> **a.** Configure R1's default FTP credentials: username **cisco**, password **cisco**.
> **b.** Use FTP to copy the file **c2900-universalk9-mz.SPA.155-3.M4a.bin** from SRV1 to R1's flash drive.
> **c.** Reboot R1 using the new IOS file, and then delete the old one from flash.

R1(config)#ip ftp username cisco
R1(config)#ip ftp password cisco
R1(config)#boot system flash:c2900-universalk9-mz.SPA.155-3.M4a.bin
R1(config)#exit

10. Configure SSH for secure remote access on all routers and switches.

> **a.** Use the largest modulus size for the RSA keys.
> **b.** Allow SSHv2 connections only.
> **c.** Create standard ACL 1, only allowing packets sourced from Office A's PCs subnet. Apply the ACL to all VTY lines to restrict SSH access.
> **d.** Allow only SSH connections to the VTY lines.
> **e.** Require users to log in with a local user account when connecting via SSH.
> **f.** Configure synchronous logging on the VTY lines.

R1(config)#crypto key generate rsa
R1(config)#ip ssh version 2
R1(config)#access-list 1 permit 10.0.1.0 0.0.0.255
R1(config)#line vty 0 15
R1(config-line)#access-class 1 in
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#logging synchronous
R1(config-line)#exit

11. Configure static NAT on R1 to enable hosts on the Internet to access SRV1 via the IP address **203.0.113.113**.
R1(config)#ip nat inside source static 10.5.0.4 203.0.113.133
R1(config)#interface range g0/0/0,g0/1/0
R1(config-if-range)#ip nat outside
R1(config-if-range)#ip range g0/0-1
R1(config-if-range)#ip nat inside
R1(config-if-range)#exit

12. Configure pool-based dynamic PAT on R1 to enable hosts in the Office A PCs, Office A Phones, Office B PCs, Office B Phones, and Wi-Fi subnets to access the Internet.

a. Use standard ACL 2 to define the appropriate inside local address ranges in the following order:

i. Office A PCs: 10.1.0.0/24
ii. Office A Phones: 10.2.0.0/24
iii. Office B PCs: 10.3.0.0/24
iv. Office B Phones: 10.4.0.0/24
v. Wi-Fi: 10.6.0.0/24

R1(config)#ip access-list standard 2
R1(config-std-nacl)#permit 10.1.0.0 0.0.0.255
R1(config-std-nacl)#permit 10.2.0.0 0.0.0.255
R1(config-std-nacl)#permit 10.3.0.0 0.0.0.255
R1(config-std-nacl)#permit 10.4.0.0 0.0.0.255
R1(config-std-nacl)#permit 10.6.0.0 0.0.0.255

b. Define a range of inside global addresses called **POOL1**, specifying the range 203.0.113.200 to 203.0.113.207 with a /29 netmask.
R1(config)#ip nat pool POOL1 203.0.113.200 203.0.113.207 netmask 255.255.255.248

c. Map ACL 2 to POOL1 and enable PAT. Confirm that hosts can access the Internet by pinging jeremysitlab.com.
R1(config)#ip nat inside source list 2 pool POOL1 overload

13. Disable CDP on all devices and enable LLDP instead.

a. Disable LLDP Tx on each Access switch's access port (F0/1).

R1**(config)**#no cdp run
R1**(config)**#lldp run

ASW-A1**(config)**#no cdp run
ASW-A1**(config)**#lldp run
ASW-A1**(config)**#interface f0/1
ASW-A1**(config-if)**#no lldp transmit

Part 7 – Security: ACLs and Layer-2 Security Features

1.Configure extended ACL **OfficeA_to_OfficeB** where appropriate:

a. Allow ICMP messages from the **Office A PCs** subnet to the **Office B PCs** subnet.
b. Block all other traffic from the **Office A PCs** subnet to the **Office B PCs** subnet.
c. Allow all other traffic.
d. Apply the ACL according to general best practice for extended ACLs.

DSW-A1(config)#ip access-list extended OfficeA_to_OfficeB
DSW-A1(config-ext-nacl)#permit icmp 10.0.0.0 0.0.0.255 10.3.0.0 0.0.0.255
DSW-A1(config-ext-nacl)#deny ip 10.0.0.0 0.0.0.255 10.3.0.0 0.0.0.255
DSW-A1(config-ext-nacl)#permit ip any any
DSW-A1(config-ext-nacl)#exit
DSW-A1(config)#interface vlan 10
DSW-A1(config-if)#ip access-group OfficeA_to_OfficeB in
DSW-A1(config-if)#end

2. Configure Port Security on each Access switch's F0/1 port:

a. Allow the minimum necessary number of MAC addresses on each port.
- **i.** SRV1 does not use virtualization, so it uses a single MAC address.

b. Configure a violation mode that blocks invalid traffic without affecting valid traffic. The switches should send notifications when invalid traffic is detected.
c. Switches should automatically save the secure MAC addresses they learn to the running-config.

ASW-A2(config)#interface f0/1
ASW-A2(config-if)#switchport port-security
ASW-A2(config-if)#switchport port-security maximum 2
ASW-A2(config-if)#switchport port-security violation restrict
ASW-A2(config-if)#switchport port-security mac-address sticky

3. Configure DHCP Snooping on all Access switches.

a. Enable it for all active VLANs in each LAN.
b. Trust the appropriate ports.
c. Disable insertion of DHCP Option 82.
d. Set a DHCP rate limit of 15 pps on active untrusted ports.
e. Set a higher limit (100 pps) on ASW-A1's connection to WLC1.

ASW-A1(config)# ip dhcp snooping
ASW-A1(config)# ip dhcp snooping vlan 10,20,40,99
ASW-A1(config)# no ip dhcp snooping information option
ASW-A1(config)# interface range g0/1-2
ASW-A1(config-if)# ip dhcp snooping trust
ASW-A1(config)# interface f0/1
ASW-A1(config-if)#ip dhcp snooping limit rate 15
ASW-A1(config)# interface f0/2
ASW-A1(config-if)#ip dhcp snooping limit rate 100

4.Configure DAI on all Access switches.

a. Enable it for all active VLANs in each LAN.
b. Trust the appropriate ports.
c. Enable all optional validation checks.

ASW-A1(config)# ip arp inspection vlan 10,20,40,99
ASW-A1(config)# ip arp inspection validate dst-mac src-mac ip
ASW-A1(config)# interface range g0/1-2
ASW-A1(config-if)# ip arp inspection trust

Part 8 – IPv6

1. To prepare for a migration to IPv6, enable IPv6 routing and configure IPv6 addresses on R1, CSW1, and CSW2:

a. R1 G0/0/0: 2001:db8:a::2/64
b. R1 G0/1/0: 2001:db8:b::2/64
c. R1 G0/0 and CSW1 G1/0/1: Use prefix 2001:db8:a1::/64 and EUI-64 to generate an interface ID for each interface.
d. R1 G0/1 and CSW2 G1/0/1: Use prefix 2001:db8:a2::/64 and EUI-64 to generate an interface ID for each interface.
e. CSW1 Po1 and CSW2 Po1: Enable IPv6 without using the 'ipv6 address' command.

R1(config)#ipv6 unicast-routing
R1(config)#interface g0/0/0
R1(config-if)#ipv6 address 2001:db8:a::2/64
R1(config)#interface g0/1/0
R1(config-if)#ipv6 address 2001:db8:b::2/64
R1(config)#interface g0/0
R1(config-if)#ipv6 address 2001:db8:a1::/64 eui-64
R1(config)#interface g0/1
R1(config-if)#ipv6 address 2001:db8:a2::/64 eui-64

2. Configure two default static routes on R1:

a. A recursive route via next hop 2001:db8:a::1.
b. A fully-specified route via next hop 2001:db8:b::1. Make it a floating route by configuring the AD 1 higher than default.

R1(config)#ipv6 route ::/0 2001:db8:a::1
R1(config)#ipv6 route ::/0 2001:db8:b::1 2

Part 9 – Wireless

1.Access the GUI of WLC1 (https://10.0.0.7) from one of the PCs.

a. Username: admin
b. Password: adminPW12

2.Configure a dynamic interface for the Wi-Fi WLAN (10.6.0.0/24)

a. Name: Wi-Fi
b. VLAN: 40
c. Port number: 1
d. IP address: .4 of its subnet
e. Gateway: .1 of its subnet
f. DHCP server: 10.0.0.76

3. Configure and enable the following WLAN:

a. Profile name: Wi-Fi
b. SSID: Wi-Fi
c. ID: 1
d. Status: Enabled
e. Security: WPA2 Policy with AES encryption, PSK of cisco123